Reviewer_Status_Report

Generated 2026-09-08

Contents

Reviewer Response Status — IEEE Access 33821

> 12 comments, one-shot resubmission. Every item below states what the reviewer asked, what changed, where the evidence lives, and what (if anything) is still open. Manuscript sources: `paper-access.tex` (A) and `paper-build.tex` (B), kept in sync. Work line: `main`.

At a glance

Comment	Status	Remaining
R1-1 single corpus, N=31	In manuscript	Owner read-through
R1-2 complex-device pins	In manuscript	Owner read-through

Comment	Status	Remaining
R1-3 manual thresholds	In manuscript	Nothing
R1-4 no OCR, not simulatable	In manuscript	Nothing (PDF rebuilds it)
R1-5 small circuits	In manuscript	Nothing
R1-6 memristor citations	Done	Template check
R2-1 statistics, roadmap	In manuscript	Owner read-through
R2-2 VLM oracle, cost	In manuscript	Nothing
R2-3 SPICE title	Done	PDF rebuild (T7)
R2-4 crossover merges	Partial by design	Nothing possible without new data
R2-5 ablations	Partial	Nothing further provable
R2-6 single-scalar limit	Closed	Nothing

Shared language enforced throughout: nonsignificance is not equivalence; no global short-free guarantee; component-pair F1 does not certify pin topology; GT-box results do not prove end-to-end dominance.

R1-1 — Single corpus, N=31 — In manuscript

Asked: 31 images from one corpus cannot establish generalization. **Done:** Drafter groups presented as descriptive within-corpus results with direct/inferred provenance disclosed (16 direct, 15 inferred); no held-out-drafter experiment claimed; future stratified-expansion protocol described as future, not completed. Join micro-F1 0.890 reported with its 95% bootstrap CI [0.855, 0.924]. **Remaining:** owner read-through (#85).

R1-2 — Complex-device pins — In manuscript

Asked: which complex devices are actually handled, and how? **Done:** new capability table (`tab:capabilities`) separating evaluated connectivity from generic pin construction and illustrative export, with a metric-boundary row. Switch wording made exact: shorter OBB-edge midpoint pins, long-axis AABB fallback, fixed 0.001-ohm emission only when pins 0 and 1 reach distinct nodes. Verified against `netlist.py` / `spice.py`. **Remaining:** owner read-through (#83).

R1-3 — Manual thresholds — In manuscript

Asked: fixed scale-relative parameters (reviewer names alpha explicitly). **Done:** multipliers stated as fixed and manual (0.62/0.30/0.20, clamps 24–60/11–28/8–20);

directional scoring fixed at alpha 0.35, fallback-only, with the exact formula; completion reach at 4x clamped pin scale; reach sweep reported as macro 0.895–0.903; extremes (rescaling, heterogeneous symbol sizes) conceded untested. Method prose rewritten from code (`join_graph.py`, `completion.py`). **Remaining:** nothing.

R1-4 — No OCR, not simulatable — In manuscript

Asked: no component values are read, so output cannot simulate. **Done:** scope sentences in abstract, pipeline overview, metric definition, and conclusion; retitle to Structural Circuit Netlists (title, running heads, diagram node); export labeled illustrative throughout, with external values/models required. **Remaining:** nothing (PDF rebuild carries it).

R1-5 — Small circuits — In manuscript

Asked: narrow coverage (3–14 supported components, median 7). **Done:** dense-bus/multilayer limitation stated; histogram caption gives the true distribution (15 ≤5, 12 ≥10, median 7, range 3–14); no generalization beyond the tested population claimed. **Remaining:** nothing (builder to reconcile histogram label during T7).

R1-6 — Memristor citations — Done

Asked: cite two memristor-circuit papers. **Done:** both records abstract-verified (IEEE TCE Feb 2026; IEEE TII vol. 22, 2026) and cited in Related Work as neighboring SPICE-deployment domains, with bibliography entries in both sources. Response flipped from decline to cited. **Remaining:** template structure check (#84).

R2-1 — Statistics, roadmap — In manuscript

Asked: statistical treatment and research roadmap. **Done:** four-strata sampling description, independent annotation/adjudication, effort and provenance logs; "same accuracy" language fixed; historical timing honestly absent; roadmap paragraph added. **Remaining:** owner read-through (#85).

R2-2 — VLM comparison — In manuscript

Asked: fairness and cost framing of the VLM reference. **Done:** VLM recast as an oracle-component-box diagnostic (both methods get annotated boxes); paired diff +0.033, CI [−0.009, +0.078] reported as nonsignificant, never as equivalence; cost multiples and superiority claims removed. **Remaining:** nothing.

R2-3 — SPICE title — Done

Asked: title overclaims simulation-ready SPICE output. **Done:** retitled end-to-end (title, headers, abstract, conclusion, figure node) with the metric boundary stated: the metric measures component-pair connectivity, not pin correctness or simulation equivalence. **Remaining:** PDF rebuild (T7).

R2-4 — Crossover merges — Partial by design

Asked: effect of crossing wires on netlist correctness, including detector misses. **Done:** causal two-edge C242 intervention reported with full tradeoff (TP/FP/FN 27/4/0 → 19/0/8, artifact `crossover-causal.json`); stated plainly that removing edges also removes true connections (no mitigation-gain claim) and that detector-miss effects were not measured. **Remaining:** nothing achievable without new experiments — submitted as an honest partial.

R2-5 — Ablations — Partial

Asked: evidence for each module's contribution ("severely insufficient"). **Done:** genuine base/full table; fixed-pixel base 0.820 vs scale-relative 0.816 reported as observed (not spun); full-pipeline ties at 0.890 presented as bounded negative evidence, never as proof that mechanisms are dispensable; missing full-occlusion and end-to-end scale ablations disclosed in paper and response. **Remaining:** nothing further provable on existing data.

R2-6 — Single-scalar limit — Closed

Asked: discussion-only point on evaluation limits. **Done:** disclosure kept as-is; no change needed and none made.

Reference numbers

Join 0.890 (418/37/66) on all 31; perfect-wire 0.88983 (420/40/64). Wire ledger: 0.9755 = 10°/18px config, 0.9726 = 12°/8px production. Otsu 0.789. VLM 0.923, 21/31 exact. Reach sweep macro 0.895–0.903. Numeric gate: 192/192 stored comparisons pass (SHAs in the amended plan).